**Math 510** **Fall 2026**
**Homework 5** **Drew Armstrong**

**1. Evaluating Polynomials at Matrices.** Given a polynomial $f(x) = \alpha_0 + \alpha_1 x + \cdots + \alpha_d x^d \in \mathbb{C}[x]$ and a square matrix $A \in \mathbb{C}^{n\times n}$, we define the matrix

$$f(A) := \alpha_0 I_n + \alpha_1 A + \alpha_2 A^2 + \cdots + \alpha_d A^d \in \mathbb{C}^{n\times n}.$$

Note that for diagonal matrices we have

$$f\begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} = \begin{pmatrix} f(\lambda_1) & & \\ & \ddots & \\ & & f(\lambda_n) \end{pmatrix}.$$

(a) If $f(A) = O$ is the zero matrix, prove that every eigenvalue $\lambda \in \mathbb{C}$ of $A$ satisfies $f(\lambda) = 0$.
(b) If $A = CBC^{-1}$, prove that $f(A) = Cf(B)C^{-1}$.

Both properties also hold for power series $f(x)$, assuming convergence.

**2. Dynamical Systems.** Consider a square matrix $A \in \mathbb{R}^{n\times n}$. Suppose that a sequence of points $\mathbf{x}_0, \mathbf{x}_1, \ldots \in \mathbb{R}^n$ is defined recursively by $\mathbf{x}_k := A\mathbf{x}_{k-1}$. Then by induction we have $\mathbf{x}_k = A^k\mathbf{x}_0$. On the other hand, suppose that $\mathbf{x}(t) = (x_1(t), \ldots, x_n(t))$ is a continuously moving point in $\mathbb{R}^n$ with velocity vector $\mathbf{x}'(t) = (x_1'(t), \ldots, x_n'(t))$. Suppose that the velocity is equal to $A$ times the position at all times, i.e., suppose that $\mathbf{x}'(t) = A\mathbf{x}(t)$ for all $t$. Then we must have $\mathbf{x}(t) = \exp(tA)\mathbf{x}(0)$. Consider the following explicit matrix $A$ and initial conditions $\mathbf{x}_0, \mathbf{x}(0)$:

$$A = \frac{1}{6}\begin{pmatrix} 5 & 4 \\ 2 & -2 \end{pmatrix}, \qquad \mathbf{x}_0 = \mathbf{x}(0) = \begin{pmatrix} 2 \\ 5 \end{pmatrix}.$$

Explicitly diagonalize the matrix $A$ and use this to find explicit formulas for $\mathbf{x}_k$ and $\mathbf{x}(t)$. [Hint: If $A = X\Lambda X^{-1}$ then from 1(b) we have $A^k = X\Lambda^k X^{-1}$ and $\exp(tA) = X\exp(t\Lambda)X^{-1}$.]

**3. Idempotent Matrices.** Let $P \in \mathbb{C}^{n\times n}$ satisfy $P^2 = P$ and have rank $0 < r \leq n$. We will show that $P = AB^*$ for some (non-unique) matrices $A, B \in \mathbb{C}^{n\times r}$ satisfying $B^*A = I_r$.

(a) Show that the eigenvalues of $P$ are in the set $\{0, 1\}$. [Hint: Use 1(a).]
(b) Let $E_0$ and $E_1$ be the 0- and 1-eigenspaces of $P$.[1] Prove that

$$\mathbb{C}^n = E_0 \oplus E_1.$$

[Hint: For any $\mathbf{x} \in \mathbb{C}^n$ we can write $\mathbf{x} = (\mathbf{x} - P\mathbf{x}) + P\mathbf{x}$. Show that $\mathbf{x} - P\mathbf{x} \in E_0$ and $P\mathbf{x} \in E_1$.]
(c) Since $\operatorname{rank}(P) = r$, we have $\dim E_1 = r$. Let $\mathbf{x}_1, \ldots, \mathbf{x}_r$ be a basis for $E_1$ and let $\mathbf{x}_{r+1}, \ldots, \mathbf{x}_n$ be a basis for $E_0$, so that $\mathbf{x}_1, \ldots, \mathbf{x}_n \in \mathbb{C}^n$ is a basis for $\mathbb{C}^n$. Let $X \in \mathbb{C}^{n\times n}$ be the matrix with columns $\mathbf{x}_1, \ldots, \mathbf{x}_n$, which is invertible. Show that

$$P = X\left(\begin{array}{c|c} I_r & O_{r,n-r} \\ \hline O_{n-r,r} & O_{n-r,n-r} \end{array}\right)X^{-1}.$$

(d) Use part (c) to prove that $P = AB^*$ for some matrices $A, B \in \mathbb{C}^{n\times r}$ satisfying $B^*A = I_r$. [Hint: Let $A$ be the first $r$ columns of $X$ and let $B^*$ be the first $r$ rows of $X^{-1}$.]
(e) Suppose, in addition, that $P^* = P$. In this case show that we can choose the matrix $X$ to be unitary, so that $P = AA^*$. [Hint: For $\mathbf{x} \in E_0$ and $\mathbf{y} \in E_1$ we have $\langle \mathbf{x}, \mathbf{y}\rangle = \langle \mathbf{x}, P\mathbf{y}\rangle = \langle P^*\mathbf{x}, \mathbf{y}\rangle$.]

**4. Independence of Eigenvectors.** Let $A \in \mathbb{C}^{n\times n}$ and let $\mathbf{x}_1, \ldots, \mathbf{x}_k \in \mathbb{C}^n$ be eigenvectors corresponding to distinct eigenvalues $\lambda_1, \ldots, \lambda_k \in \mathbb{C}$. By distinct, we mean that $i \neq j$ implies $\lambda_i \neq \lambda_j$.

[1] It is possible that one of these eigenspaces is trivial.

(a) For each index $i$, consider the polynomial

$$p_i(x) = \prod_{j \neq i} \frac{x - \lambda_j}{\lambda_i - \lambda_j} \in \mathbb{C}[x].$$

Note that $p_i(\lambda_j) = \delta_{ij}$. Use this to show that $p_i(A)\mathbf{x}_j = \delta_{ij}\mathbf{x}_j$.

(b) Use part (a) to prove that $\mathbf{x}_1, \ldots, \mathbf{x}_k$ is an independent set. [Hint: Suppose $\sum_j \alpha_j \mathbf{x}_j = \mathbf{0}$. Multiply both sides by the matrix $p_i(A)$.]

**5. Positive Semidefinite Matrices.** Let $B \in \mathbb{C}^{n \times n}$ be *self-adjoint*, i.e., $B^* = B$.

(a) Show that $\langle \mathbf{x}, B\mathbf{x} \rangle \in \mathbb{R}$ for all $\mathbf{x} \in \mathbb{C}^n$. Use this to show that every eigenvalue of $B$ is real. [Hint: Show that $\langle \mathbf{x}, B\mathbf{x} \rangle^* = \langle \mathbf{x}, B\mathbf{x} \rangle$ for all $\mathbf{x} \in \mathbb{C}^n$. If $B\mathbf{x} = \lambda\mathbf{x}$ then $\langle \mathbf{x}, B\mathbf{x} \rangle = \lambda \langle \mathbf{x}, \mathbf{x} \rangle$.]

(b) Show that the following are equivalent:

1. $\langle \mathbf{x}, B\mathbf{x} \rangle \geq 0$ for all $\mathbf{x} \in \mathbb{C}^n$.
2. Every eigenvalue of $B$ is $\geq 0$.
3. $B = A^*A$ for some matrix $A$.

When any of these properties holds, we say that $B$ is *positive semidefinite.*[2]

[Hint: If $B^* = B$, the Spectral Theorem tells us that $B = U\Lambda U^*$ for some unitary matrix $U$ and diagonal matrix $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$. If $\lambda_i \geq 0$ for all $i$ then it makes sense to define $\Lambda^{1/2} := \mathrm{diag}(\sqrt{\lambda_1}, \ldots, \sqrt{\lambda_n})$ where $\sqrt{\lambda_i}$ is the unique non-negative square root of $\lambda_i$. Then let $A := \Lambda^{1/2}U^*$.]

**6. Normal Operators.** Let $V$ be a Hilbert space and let $A : V \to V$ be a continuous operator satisfying $A^*A = AA^*$. (You can think of $V = \mathbb{C}^n$, in which case $A^*$ is the conjugate transpose matrix.)

(a) Prove that $\langle A\mathbf{x}, A\mathbf{y} \rangle = \langle A^*\mathbf{x}, A^*\mathbf{y} \rangle$ for all $\mathbf{x}, \mathbf{y} \in V$.

(b) Show that $A\mathbf{x} = \mathbf{0}$ if and only if $A^*\mathbf{x} = \mathbf{0}$. [Hint: Taking $\mathbf{y} = \mathbf{x}$ in part (a) gives $\|A\mathbf{x}\| = \|A^*\mathbf{x}\|$.]

(c) Show that $A\mathbf{x} = \lambda\mathbf{x}$ if and only if $A^*\mathbf{x} = \lambda^*\mathbf{x}$, hence $A$ and $A^*$ have the same eigenvectors. [Hint: Consider the operator $B = A - \lambda I$. Show that $B^*B = BB^*$ and then use part (b).]

(d) If $A\mathbf{x} = \lambda\mathbf{x}$ and $A\mathbf{y} = \mu\mathbf{y}$ with $\lambda \neq \mu$, prove that $\langle \mathbf{x}, \mathbf{y} \rangle = 0$. [Hint: Use part (c) to show that $\lambda \langle \mathbf{x}, \mathbf{y} \rangle = \mu \langle \mathbf{x}, \mathbf{y} \rangle$.]

**?. Total Least Squares.** Maybe?

**Challenge Problem (Perron-Frobenius).** We will write $M > 0$ or $\mathbf{x} > 0$ to indicate that a matrix $M$ or vector $\mathbf{x}$ has strictly positive (real) entries. We will accept the following result without proof: *Any positive square matrix $A > 0$ has a positive eigenvector $\mathbf{v} > 0$.*

(a) Let $A > 0$ be a positive square matrix, so $A^T > 0$ is also a positive square matrix. Let $\mathbf{v} > 0$ and $\mathbf{w} > 0$ be corresponding positive eigenvectors. Show that the eigenvalue is the same, i.e., that $A\mathbf{v} = r\mathbf{v}$ and $A^T\mathbf{w} = r\mathbf{w}$ for some $r > 0$. [Hint: Consider $\mathbf{w}^T A\mathbf{v}$.] To simplify notation, we will replace $\mathbf{v}$ with $\mathbf{v}/(\mathbf{w}^T\mathbf{v})$, so that $\mathbf{w}^T\mathbf{v} = 1$. Note that this preserves the properties $A\mathbf{v} = r\mathbf{v}$ and $\mathbf{v} > 0$. We will show that

$$\left(\frac{1}{r}A\right)^k \to \mathbf{v}\mathbf{w}^T \text{ entrywise, as } k \to \infty.$$

(b) Define $P := \mathbf{v}\mathbf{w}^T$. Show that $P^2 = P$ and $AP = PA = rP$. Let $B_\varepsilon := A - \varepsilon P$, for some $\varepsilon$. Observe that $B_\varepsilon \mathbf{v} = (r - \varepsilon)\mathbf{v}$ and $\mathbf{w}^T B_\varepsilon = (r - \varepsilon)\mathbf{w}^T$. Show that

$$\left(\frac{1}{r}A\right)^k = P + \left(\frac{1}{r}B_\varepsilon\right)^k (I - P) \text{ for all } k \geq 1.$$

[2] In the case that $B$ is invertible, then $\langle \mathbf{x}, B\mathbf{x} \rangle > 0$ for all $\mathbf{x} \in \mathbb{C}^n$, every eigenvalue of $B$ is strictly positive, and $B = A^*A$ for some matrix $A$ with independent columns. In this case we say that $B$ is *positive definite*. The terminology is not great.

Our goal is to show that $(B_\varepsilon/r)^k \to O$ entrywise.

(c) For any vector $\mathbf{x} = (x_1, \ldots, x_n)$ we will use the notation $|\mathbf{x}| := (|x_1|, \ldots, |x_n|)$. For any positive matrix $M > 0$, show that

$$|M\mathbf{x}| \leq M|\mathbf{x}| \text{ entrywise.}$$

Since $A > 0$ we can choose $\varepsilon > 0$ such that $B_\varepsilon = A - \varepsilon P > 0$ is positive, and hence $B_\varepsilon^k > 0$ is positive for all $k \geq 1$. It follows that $|B^k\mathbf{x}| \leq B^k|\mathbf{x}|$ entrywise for all $\mathbf{x}$. Use this to show that

$$\mathbf{w}^T|B_\varepsilon^k\mathbf{x}| \leq \mathbf{w}^T B_\varepsilon^k|\mathbf{x}| \text{ for all } k \geq 1 \text{ and } \mathbf{x}.$$

Since $\mathbf{v} > 0$ and $B_\varepsilon\mathbf{v} = (r-\varepsilon)\mathbf{v}$, we also have $\varepsilon < r$. Use the previous inequality and the fact that $\mathbf{w}^T B_\varepsilon^k|\mathbf{x}| = (r-\varepsilon)^k\mathbf{w}^T|\mathbf{x}|$ to show that $(B/r)^k\mathbf{x} \to \mathbf{0}$ entrywise, for all $\mathbf{x}$. Show that this implies $(B/r)^k \to O$ entrywise.

(d) **Application: Markov Chains.** Let $\mathbf{1}$ be the vector of all ones. Let $M > 0$ satisfy $\mathbf{1}^T M = \mathbf{1}^T$, meaning that every column of $M$ sums to 1. In this case, show that $M^k \to \mathbf{p}\mathbf{1}^T$ for some positive vector $\mathbf{p} > 0$ satisfying $\mathbf{1}^T\mathbf{p} = 1$ and $M\mathbf{p} = \mathbf{p}$.